

3D Coordinates Extension — English Worksheet (Intermediate)

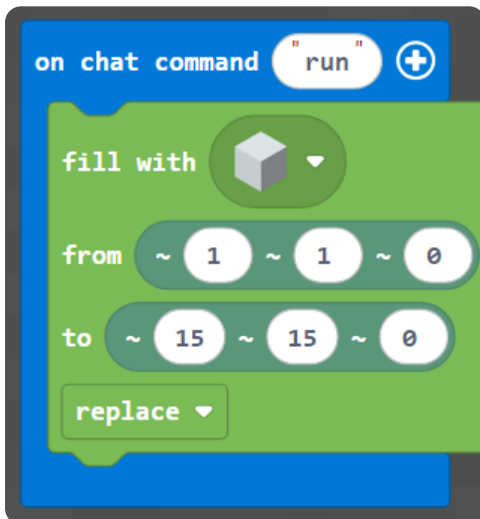
Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 3) ·

Time: about 40 minutes

A bonus picture challenge: copy a whole mob — a **pink axolotl** — onto your wall. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict 🧠

Read each set of steps. Write a sentence about what you will see.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Three pink blocks share $y = 13$. What shape do they make, and which number stays the same?


```
place black block at (8, 7)
place black block at (8, 8)
place black block at (8, 9)
```

These three share $x = 8$. What shape do they make, and which number changes?


```
place pink block at (6, 10)
place black block at (6, 10)
```

Two blocks aim for the same (x, y) . What do you see — pink, black, or both? Why?

2 · Spot the Bug

Each code block is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — This should make the axolotl's two eyes, one at (6, 10) and one at (11, 10). Both land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The dragon's eye should sit at (8, 10). The two numbers are **swapped**, so it lands in the wrong place.

```
place purple block at (10, 8)
```

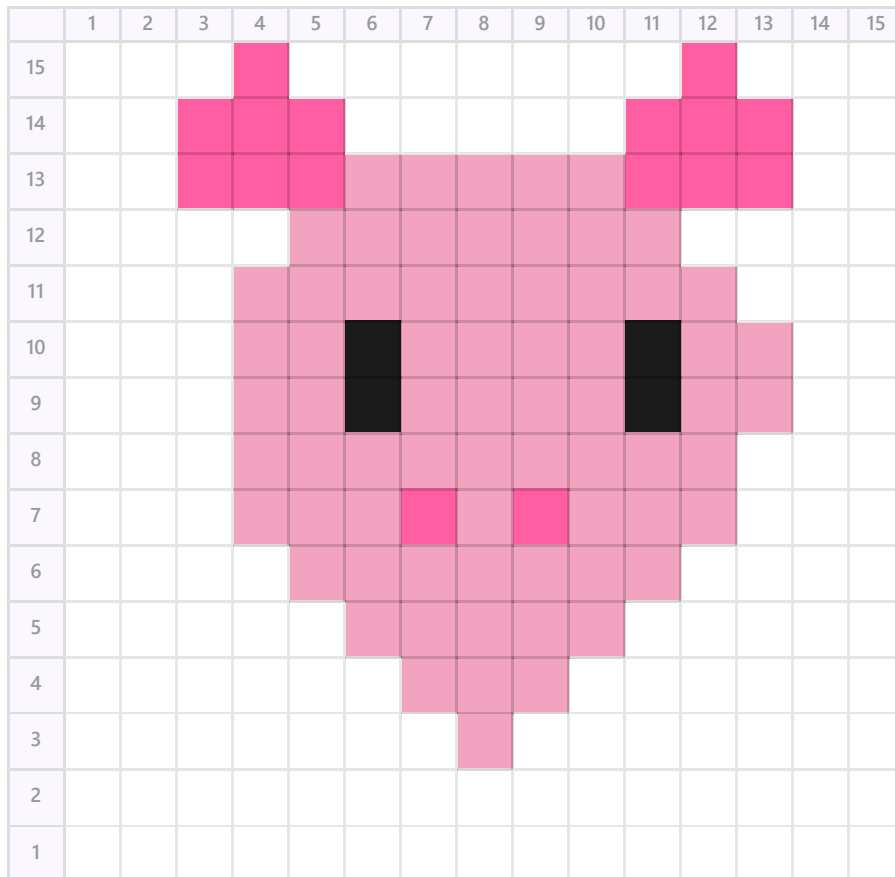
Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block). Build the 🐸 axolotl block by block, reading each colored square's (x, y) — **x** across the top, **y** down the side.

🐸 **Axolotl** (pink · #f2a3c0, frills · #ff5fa2, eyes · black)



Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.